

浙江大学

Zhejiang University-University of Illinois Urbana-Champaign Institute

Undergraduate Thesis (Design)



Thesis Title	Evaluating Memory-Augmented Agent Systems with Medical Reasoning
Student Name	Zihan Xu
Student ID	3220110781
Supervisor	Zuozhu Liu
Class of year	2026
Major	Computer Engineering
Submission date	May 22, 2026

Content Checklist

(Student should bind this form on the first page of the thesis after the cover page. The form should be filled out by the supervisor after the oral defense.)

Student Name	Zihan Xu	ZJU ID	3220110781	Major	Computer Engineering
Supervisor	Zuozhu Liu			Academic Title	Assistant Professor
Thesis title	Evaluating Memory-Augmented Agent Systems with Medical Reasoning				
Content Checklist					Evaluation Section (√)
Senior Thesis	1. Cover (Use the unified cover, blue recommended)			Good	Fair
	2. Academic integrity Commitment of Senior Thesis				
	3. Acknowledgements				
	4. Abstract				
	5. Table of Contents				
	6. Main Content (Page numbers start from the main text)				
	7. Reference				
	8. Appendix (if need)				
	9. Author's Biography				
	10. Task plan of senior thesis				
	11. Assessment Form of senior thesis				
	12. On-site Minutes Form of Oral Defense				
Signature of Supervisor:				Date:	

**ZJU-UIUC INSTITUTE**Zhejiang University-University of Illinois Urbana-Champaign Institute
浙江大学伊利诺伊大学厄巴纳香槟校区联合学院

Academic integrity Commitment of Senior Thesis

1. I solemnly promise that the submitted thesis has been completed under the guidance of my supervisor in strict accordance with the relevant regulations of the university and the college.
2. In my thesis, except for the parts specifically marked and acknowledged, it does not contain research results that have been published or written by others, nor does it contain materials used to obtain degrees or certificates from Zhejiang University or other educational institutions.
3. Any contributions made by my colleagues to this research have been clearly stated and acknowledged in the thesis.
4. I promise that there has been no fabrication of data in the process of completing the thesis.
5. If there is any infringement of intellectual property rights in this thesis, I shall bear the corresponding legal responsibilities.
6. I fully understand that Zhejiang University has the right to retain and submit copies and disks of this thesis (design) to relevant departments or agencies, and to allow the thesis (design) to be consulted and borrowed. I authorize Zhejiang University to include all or part of this thesis (design) in relevant databases for retrieval and dissemination, and to save and compile this thesis (design) by means of photocopying, reduction, or scanning.

Author's Signature:

Supervisor's Signature:

Date:

Date:

Acknowledgements

I would like to express my sincere gratitude to my supervisor, Professor Zuozhu Liu, for his guidance, patience, and insightful feedback throughout the course of this thesis project. His expertise in medical AI provided the intellectual foundation upon which this work was built.

I am grateful to my collaborators on the LongMedBench project, Yanzhen Chen and Xiaocheng Zhang, whose work on the factual question-answering and temporal reasoning components provided essential context for the long-horizon reasoning framework developed in this thesis. LongMedBench has been submitted to MICCAI 2026 and the rebuttal stage is completed recently. We thank the anonymous reviewers at the conference for providing many constructive suggestions regarding our framework.

Many thanks to the senior students of ZJUI-AI4H, especially Yuan Wang who provided us with computing resources, and Zhiting Fan who provided concrete guidance on paper writing. The collaborative environment fostered productive discussions that sharpened the research questions and methodology.

I also acknowledge the maintainers of the MIMIC-IV database, whose commitment to open clinical data access makes research of this kind possible.

Finally, I thank my family for their unwavering support throughout my undergraduate studies, and the faculty of the Computer Engineering program at ZJUI and UIUC for providing a rigorous and inspiring academic environment.

Abstract

Clinical decision-making is inherently longitudinal — clinicians must synthesize evidence across repeated hospitalizations, diagnostic tests, and evolving treatment responses spanning months or years. However, current benchmarks for medical large language model (LLM) predominantly evaluate single-encounter behavior, focusing on model knowledge, tool-use accuracy and fact retrieval rather than the cross-visit temporal integration that real clinical practice demands. To address this gap, this thesis introduces the Long-Horizon Reasoning Task of LongMedBench, a benchmark component constructed from real-world Electronic Health Record (EHR) data (MIMIC-IV) that evaluates whether LLM agents can reason over longitudinal clinical history before selecting clinical decisions or actions. We design a three-tier memory architecture operating at three granularities: Note Memory (visit-level summaries), Event Memory (fine-grained structured event sequences), and Contextual Memory (immediate visit context). Based on the memory datasets, we systematically compare memory augmentation strategies including long-context LLM, retrieval-augmented generation (RAG), and agent memory systems (Mem0). Through experiments on GPT-5-mini, DeepSeek-v3.2, and Qwen-Turbo, we find that decision-making accuracy across all models remains modest, and critically, that memory augmentation architectures provide no significant improvement over the naive long-context baseline. Decision accuracy depends primarily on the immediate clinical context, with historical information contributing negligible marginal value regardless of memory granularity or retrieval strategy. This central finding reveals a retrieval-reasoning gap: current architectures can locate relevant historical facts but cannot effectively integrate them into proactive clinical reasoning. We discuss the implications of this gap for medical AI agent design and propose directions for memory architectures that go beyond retrieval to support structured longitudinal reasoning.

Keywords: Medical AI Agents, Clinical Decision-Making, Electronic Health Records, Long-Horizon Reasoning, Retrieval-Augmented Generation, Agent Memory Systems

Content

Chapter 1. Introduction	2
1.1 The Long-Horizon Challenge in Clinical Decision-Making	2
1.2 Medical Benchmarks for Clinical AI	2
1.3 Long-Horizon Benchmarks in the General Domain	4
1.4 Memory-Augmented Architectures for Long-Horizon Interaction	5
1.5 LongMedBench	6
1.6 Research Questions	7
1.7 Contributions	7
1.8 Scope and Structure	8
Chapter 2. Methodology & Experiments	9
2.1 EHR Data Processing Pipeline	9
2.1.1 Data Preprocessing & Event Filtering	10
2.1.2 Event Stream Construction	11
2.2 Three-Tier Memory Architecture	11
2.2.1 Note Memory	11
2.2.2 Event Memory	11
2.2.3 Contextual Memory	12
2.3 Long-Horizon Reasoning Task	12
2.3.1 Task Formulation	12
2.3.2 Evaluation Metrics	13
2.4 Experimental Setup	14
2.4.1 Models	14
2.4.2 Agent Architectures	14
2.4.3 Experimental Conditions	15
2.5 Results and Analysis	15
2.5.1 Model Level Patterns	15
2.5.2 Memory Ablations: Visit Injection and Retrieval Architectures	17
Chapter 3. Discussion and Conclusion	18
3.1 Summary of Findings	18
3.2 The Retrieval-Reasoning Gap	18
3.3 Benchmark Validity and Clinical Ground Truth	19
3.4 Limitations and Future Work	20

3.5 Reproducibility	21
3.6 Conclusion	21
References	22
Appendix	25
A1 Dataset Pipeline	25
A1.1 Raw Discharge Note Example	25
A1.2 Structured Admission and Discharge Events	26
A1.3 Radiology and Imaging Event Example	27
A2 Memory Construction	27
A2.1 LLM Convert	27
A2.2 Memory Templates	28
A3 Question Generation	30
A4 Agent Configuration	30
A5 Case Study	30
Author's Biography	38
Task Plan of Thesis Project	39
On-site Minutes of Oral Defense	40
Assessment Form of Senior Thesis	41

Chapter 1. Introduction

1.1 The Long-Horizon Challenge in Clinical Decision-Making

Clinical practice is fundamentally longitudinal. Patients with chronic conditions accumulate medical histories spanning years and dozens of hospitalizations, and at each encounter, clinicians must synthesize evidence across prior diagnostic tests, imaging studies, medication adjustments, and evolving treatment responses [1]. This capacity for cross-visit temporal integration is central to clinical reasoning and represents a capability that medical AI must acquire to be clinically useful.

As large language models (LLMs) transition from static question-answering to autonomous medical agents, their ability to navigate longitudinal clinical trajectories has become a critical benchmark for clinical readiness. Yet existing evaluation benchmarks assess single-encounter behavior: tool-use accuracy in a simulated visit, diagnosis from a single symptom set, or fact retrieval from a single discharge summary. These evaluations, while valuable, do not measure the cross-visit temporal integration that real clinical practice demands.

This thesis presents the Long-Horizon Reasoning Task of **LongMedBench**, a benchmark component constructed from MIMIC-IV [2] EHR data that evaluates whether LLM agents can reason over longitudinal clinical history before selecting next-step clinical decisions. Our work bridges two previously disconnected lines of research: the design of long-horizon interaction benchmarks from the general AI domain, and the evaluation of clinical reasoning from real-world EHR data. Through a three-tier memory architecture and systematic ablation of memory strategies, we investigate a central question: given long-context windows, RAG, and agent memory systems, how well can contemporary LLMs integrate longitudinal clinical history to inform next-step decisions?

1.2 Medical Benchmarks for Clinical AI

Medical AI evaluation has evolved from static QA to interactive agent settings, as illustrated in Table 1.1. Early benchmarks such as MedBench [3] and PromptCBLUE

Table 1.1 Medical AI benchmarks across EHR grounding, agent interaction, longitudinal design, and memory evaluation. *Legend:* EHR = real electronic health record data. Agent = interactive agent-based evaluation. Memory Study = systematic memory architecture ablation.

Benchmark	EHR	Multi-Visit	Agent	Decision-Making	Temporal	Memory Study	Year	Data Source
MedBench [3]							2023	Exams
EHRSQL [4]	✓						2022	MIMIC-IV
MedAlign [5]	✓			✓			2024	STARR
MedAgentBench [6]	✓		✓	✓			2025	STARR
AgentClinic [7]	✓		✓	✓			2025	Simulated
ER-REASON [8]	✓		✓	✓			2025	MIMIC-IV
FHIR-AgentBench [9]	✓		✓		✓		2025	MIMIC-IV
MedAgentGym [10]	✓		✓	✓			2025	Multiple
TIMER [11]	✓	✓	✓	✓	✓		2025	STARR
EHR-Bench [12]	✓	✓	✓	✓			2025	MIMIC-IV
LongMedBench	✓	✓	✓	✓	✓	✓	2026	MIMIC-IV

[13] tested LLMs’ factual knowledge recall from medical licensing exams. Second-generation work introduced real clinical data: EHRSQL [4] pioneered text-to-SQL on structured EHR; FHIR-AgentBench [9] extended this to interoperable FHIR standards; MedAlign [5] provided clinician-generated instruction-following tasks grounded in EHR data.

Both categories above assess static model capabilities; however, real-world medical tasks demand interactive reasoning. A series of agent-based evaluation frameworks has been developed, wherein researchers construct simulated environments that enable LLMs to interactively complete complex tasks. MedAgentBench [6] introduced a virtual EHR environment with 300 clinician-written tasks; AgentClinic [7] added multimodal clinical encounters; ER-REASON [8] focused on emergency room reasoning with structured subtasks; and MedAgentGym [10] provided a scalable training environment for biomedical reasoning.

Despite this progress, a critical gap persists: all current medical benchmarks evaluate single-encounter behavior. Real clinical care spans dozens of visits over years, and a patient’s current presentation cannot be understood in isolation from their clinical trajectory. Furthermore, no existing medical benchmark systematically compares memory architectures for AI agents in clinical decision-making.

1.3 Long-Horizon Benchmarks in the General Domain

In parallel with medical AI, a rich set of long-horizon benchmarks has emerged in the broader AI agent community, as shown in Table 1.2. Their task design principles inspire the construction of longitudinal clinical benchmarks, which include multi-turn interaction, cross-session memory, and progressive information revelation.

Long-Context Retrieval. RULER [14] systematically evaluates the effective context window of LLMs through synthetic tasks such as needle-in-a-haystack retrieval and variable tracking at controlled lengths up to 128K tokens. NarrativeQA [15] tests reading comprehension over full-length books and movie scripts, requiring models to answer questions that depend on information distributed across long documents. HotpotQA [16] requires reasoning across multiple documents to answer questions.

Multi-Hop and Multi-Turn Benchmarks. Beyond basic retrieval, LoCoMo [17] and LongMemEval [18] use simulated multi-session dialogues to evaluate whether models can recall and reason about information shared across temporally separated interactions. Both introduce the concept of multi-hop reasoning, which parallels the clinical need to connect evidence across visits.

Interactive Agent Benchmarks. TRIP-Bench [19] evaluates agents on executing long-horizon interactive tasks in real-world scenarios, where agents must plan, execute, and adapt across extended interactions. AMA-Bench [20] benchmarks long-horizon memory for agentic applications, measuring how well agents retain and leverage information across prolonged task sequences. In both cases, the agent navigates a sequence of steps informed by all prior history, and evaluation centers on whether historical information is effectively integrated into next-step decisions. This paradigm inspires LongMedBench’s clinical long-horizon reasoning setting.

Table 1.2 General-domain benchmarks for long-context, multi-turn, and long-horizon agent evaluation. *Legend:* Long Context = requires processing contexts exceeding typical window sizes. Interactive = involves multi-step agent-environment interaction.

Benchmark	Multi-Turn	Long Context	Interactive	Reasoning	Key Design	Year
NarrativeQA [15]		✓			Full-book comprehension	2017
HotpotQA [16]				✓	Multi-doc reasoning	2018
RULER [14]		✓			Synthetic context stress test	2024
LoCoMo [17]	✓	✓	✓	✓	Multi-session dialogues	2024
LongMemEval [18]	✓	✓	✓	✓	Long-term chat memory	2024
TRIP-Bench [19]	✓	✓	✓	✓	Real-world interactive planning	2026
AMA-Bench [20]	✓	✓	✓	✓	Memory for agentic applications	2026

1.4 Memory-Augmented Architectures for Long-Horizon Interaction

To address the challenges exposed by long-horizon benchmarks, as shown in Table 1.3, a parallel line of research has developed memory-augmented architectures that extend LLM capabilities beyond their native context windows. The motivation is straightforward: an LLM’s parameters are fixed at inference time, and its input context is finite. Without an external memory mechanism, information from prior documents, sessions, or patient visits either disappears or must be repeatedly reinserted into the prompt.

Retrieval-Augmented Generation (RAG) The foundational RAG paradigm [21] addresses this limitation by retrieving relevant documents from an external knowledge base at inference time and prepending them to the context window. In this formulation, retrieval functions as a read-only external memory: the model does not need to store all knowledge in its parameters or input prompt, but can condition generation on a small set of retrieved passages. This design has become a standard approach for factual grounding and domain adaptation, especially when the relevant knowledge base is large or frequently updated.

Agent Memory Systems However, long-horizon interaction requires more than one-shot retrieval from a static corpus. A personalized agent must preserve continuity across sessions, track evolving goals and states, update earlier assumptions, and selectively use prior experience when making future decisions [22]. MemGPT [23] introduced an OS-inspired memory hierarchy with working and archival storage, making memory management an explicit part of agent operation. Mem0 [24] provides a persistent, evolvable memory layer with add, search, and update operations. More recent frameworks push this idea further: A-MEM [25] models agentic memory decisions, where the agent autonomously determines what to retain; EverMemOS [26] proposes a self-organizing memory operating system for structured long-horizon reasoning; and Memory as Action [27] reframes memory management as an action the agent takes, integrating it into the decision loop rather than treating it as preprocessing. We adopt Mem0 [24] as the representative agent memory system in our experiments because it has achieved state-of-the-art performance on both the LoCoMo [17] and LongMemEval [18] datasets, and its clean API and production-ready design are well-suited for systematic ablation across memory granularities.

Table 1.3 Representative memory-augmented architectures for long-horizon LLM agents. *Legend:* Dynamic = memory state evolves across interactions.

Framework	Strategy	Mechanism	Dynamic	Year	Innovation
RAG [21]	Retrieve-then-generate	Embed + Retrieve		2020	External knowledge access
MemGPT [23]	Hierarchical memory	Core + Archival paging	✓	2023	OS-inspired management
Mem0 [24]	Agent memory layer	Add + Search + Update	✓	2024	Multi-session persistence
A-MEM [25]	Agentic memory	Autonomous retention	✓	2025	Agent-driven memory
Memory as Action [27]	Memory-as-decision	Context curation loop	✓	2025	Memory in decision loop
EverMemOS [26]	Self-organizing OS	Structured long-horizon	✓	2026	Self-organizing memory

1.5 LongMedBench

LongMedBench is positioned at the intersection of the two fields surveyed above. It inherits the domain grounding of medical benchmarks while adopting the task design

methodology of general long-horizon benchmarks.

1. **Innovation in medical AI:** LongMedBench is the first benchmark to combine real EHR data with interactive agent evaluation and systematic memory architecture ablation. No prior medical benchmark (Table 1.1) has systematically compared how different memory strategies—naive long-context, retrieval-augmented, and agent-managed memory—affect clinical decision-making.
2. **Innovation in long-horizon AI:** LongMedBench is the first benchmark to apply long-horizon interaction task design to real-world clinical data. Unlike general-domain benchmarks that use synthetic or curated text, LongMedBench tests long-horizon reasoning on noisy, real-world clinical trajectories with temporal structure and clinical consequence.

1.6 Research Questions

Specifically, focusing on the Long-Horizon Reasoning Task within LongMedBench, this thesis investigates three questions:

1. **Memory granularity:** Does the format of historical information affect clinical decision quality?
2. **Memory architecture:** Do RAG and agent memory systems (Mem0) improve longitudinal decision-making relative to providing the full history in context?
3. **Historical depth:** Does decision performance plateau, improve, or degrade as more historical visits are included?

1.7 Contributions

Within the broader LongMedBench project, this thesis makes four contributions:

1. **EHR-to-Agent Data Pipeline.** A reproducible pipeline transforming raw MIMIC-IV [2] records into longitudinal event streams for interactive agent evaluation,

yielding 355 patients and 6,999 hospitalization visits (mean 19.72 visits/patient, 44.91 medical events/visit).

2. **Three-Tier Memory Architecture.** A memory framework operating at three granularities, enabling systematic ablation of memory effects on decision quality: Note Memory $\mathcal{M}_N^{(i,j)}$ (coarse-grained visit summaries), Event Memory $\mathcal{M}_E^{(i,j)}$ (fine-grained structured event sequences), and Context Memory $\mathcal{M}_C^{(j,T)}$ (immediate visit context).
3. **Long-Horizon Reasoning Task.** An interactive evaluation protocol where the agent receives a partial patient trajectory through controlled memory, writes an explicit rationale, and then predicts the next clinical action or decision from pre-defined options, with ground truth extracted from actual clinical records.
4. **Empirical Finding.** Through experiments across GPT-5-mini [28], DeepSeek-v3.2 [29], and Qwen-Turbo [30] comparing Long-Context LLM, RAG, and Mem0 [24] strategies, we find that memory augmentation provides no significant improvement for longitudinal clinical decision-making. Decision accuracy depends primarily on immediate context, revealing a retrieval-reasoning gap that challenges current memory-centric approaches.

1.8 Scope and Structure

This thesis focuses exclusively on the Long-Horizon Reasoning Task on LongMedBench. The Fact-based QA and Temporal Reasoning tasks are covered in collaborators' separate theses.

Chapter 2 presents the data processing pipeline, three-tier memory architecture, long-horizon reasoning task design, experimental setup, and results. Chapter 3 discusses the implications of our findings, acknowledges limitations, and outlines future research directions.

Chapter 2. Methodology & Experiments

This chapter presents the LongMedBench long-horizon reasoning pipeline: from raw EHR data to automatically constructed decision queries for interactive agent evaluation. Section 2.1 describes the data processing pipeline. Section 2.2 presents the three-tier memory architecture. Section 2.3 formulates the Long-Horizon Reasoning Task as a template-driven clinical reasoning problem. Section 2.4 details the experimental design. Section 2.5 presents results and analysis.

2.1 EHR Data Processing Pipeline

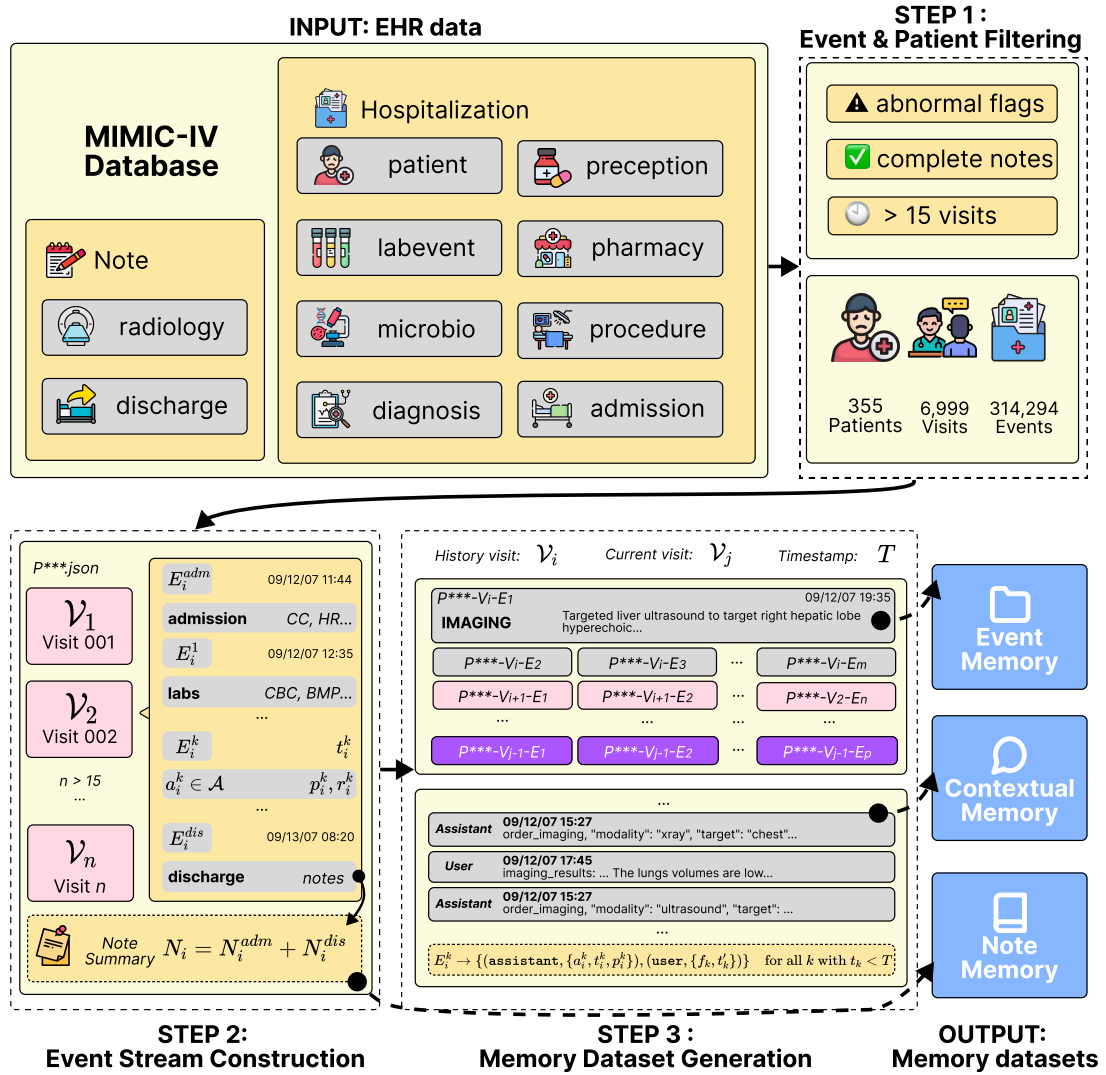


Figure 2.1 LongMedBench EHR data processing pipeline from MIMIC-IV records to patient event streams.

LongMedBench is constructed from MIMIC-IV v3.1 [2], a publicly available de-identified database containing comprehensive clinical data for over 100,000 patients at Beth Israel Deaconess Medical Center (2008–2022). To convert this database into an interactive agent-compatible environment, we design a three-stage pipeline (Figure 2.1).

2.1.1 Data Preprocessing & Event Filtering

The preprocessing stage converts EHR records of MIMIC-IV [2] into the longitudinal substrate used by the memory architecture. We retain clinically salient structured events from the `hosp` module, including abnormal laboratory results, microbiology, medications, diagnoses, procedures, and pharmacy records. Each hospitalization is anchored by an admission event and a discharge event, with matched discharge summaries from the `note` module. Meanwhile, radiology reports from the `note` module are parsed as imaging events for downstream memory construction. This filtering keeps the data representation compact enough for controlled evaluation while preserving the information needed to compare memory granularity.

Resulting Cohort. The filtered dataset comprises 355 patients with 6,999 inpatient visits. Mean visits per patient: 19.72 (median = 18.00, SD = 5.72). Mean medical events per visit: 44.91 (median = 25.00, SD = 70.32). The distribution of data is shown in Figure 2.2. This dense multi-session structure provides the longitudinal basis for evaluating memory retention and temporal reasoning.

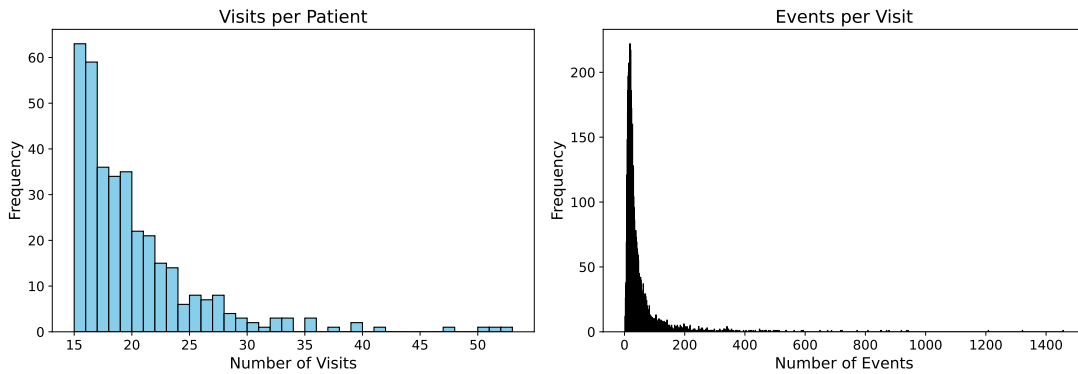


Figure 2.2 Visit-count distribution in the filtered LongMedBench cohort.

2.1.2 Event Stream Construction

We transform filtered records into a unified, time-ordered event stream for each patient. A medical event is any discrete clinical occurrence extracted from the structured hospitalization data. Formally, the k th event in $\mathcal{V}_i$ is $E_i^k := \{a_i^k, t_i^k, p_i^k, o_i^k\}$, where a_i^k is the action type from the clinical action space $\mathcal{A}$, *including imaging, lab tests, medication, etc.*, t_i^k is the timestamp, p_i^k is the action parameter (*e.g., CT, X-ray, etc. for imaging*), and o_i^k is the observation result (*e.g., a radiology report*).

We define a visit as the period spanning from an admission event E_i^{adm} to the corresponding discharge event E_i^{dis} . All other structured events occurring between these two timestamps are collected and ordered chronologically. Formally, each visit $\mathcal{V}_i$ can be represented as $\mathcal{V}_i = \{E_i^{adm}, E_i^1, E_i^2, \dots, E_i^m, E_i^{dis}\}$. The complete patient stream can therefore be denoted as $\mathcal{S}_P = \{\mathcal{V}_1, \mathcal{V}_2, \dots, \mathcal{V}_n\}$, where n is the total number of visits.

2.2 Three-Tier Memory Architecture

To provide the agents with controlled provision of historical information, we define a three-tier memory architecture that systematically varies the granularity of historical information. Given the earliest historical visit $\mathcal{V}_i$, the current visit $\mathcal{V}_j$, and the current timestamp T , the agent’s memory access is strictly bounded to prevent event leakage.

2.2.1 Note Memory

$\mathcal{M}_N^{(i,j)} := \{N_i, N_{i+1}, \dots, N_{j-1}\}$, which contains continuous visit-level discharge summaries from $\mathcal{V}_i$ to $\mathcal{V}_{j-1}$. Note Memory is compact and preserves clinical narrative structure, emphasizing the global trajectory, but discards fine-grained temporal and event-level details.

2.2.2 Event Memory

$\mathcal{M}_E^{(i,j)} := \bigcup_{k=i}^{j-1} \mathcal{V}_k$, which contains all clinical events from $\mathcal{V}_i$ to $\mathcal{V}_{j-1}$. Events are flattened into a single chronological sequence, discarding visit-level boundaries and forming a unified trajectory. Event Memory is information-complete, and its volume creates a natural stress test for both long-context capabilities and retrieval strategies.

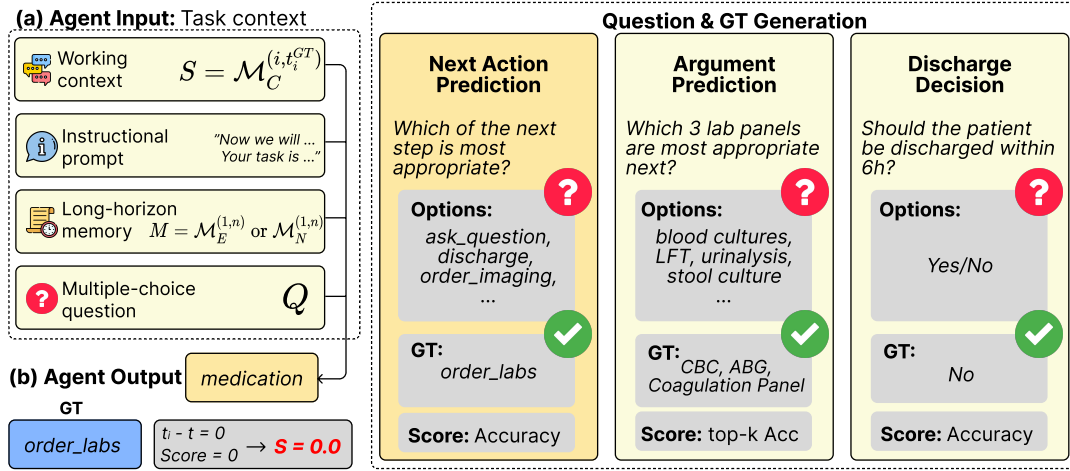


Figure 2.3 Long-Horizon Reasoning Task interface for memory-augmented clinical agents.

2.2.3 Contextual Memory

$\mathcal{M}_C^{(j, T)} := \{(r_0, m_0), (r_1, m_1), \dots, (r_P, m_P)\}$, where r_k and m_k denote the dialog role (assistant or user) and message content, respectively. Through an LLM-based rewriting template, any event E_j^k in $\mathcal{V}_j$ with timestamp $t_j^k < T$ is transformed into a conversation trajectory between the agent and the clinical environment, simulating the agent's immediate interaction state up to the decision point.

Contextual Memory represents the baseline information always available to the agent at the current interaction state. Historical memory ($\mathcal{M}_N^{(i, j)}$ or $\mathcal{M}_E^{(i, j)}$) is the experimental variable, enabling isolation of the marginal contribution of history beyond the immediate clinical context.

2.3 Long-Horizon Reasoning Task

2.3.1 Task Formulation

For a sampled ground-truth event E_i^{GT} in the current visit $\mathcal{V}_i$ with timestamp t_i^{GT} , the working context is defined as $S = \mathcal{M}_C^{(i, t_i^{GT})}$, containing all interaction history available before the target event. To ensure sufficient historical trajectory length, we only sample rounds where $n \geq 15$. The long-horizon memory is denoted as $M = \mathcal{M}_E^{(1, n)}$ or $M = \mathcal{M}_N^{(1, n)}$, depending on whether Event Memory or Note Memory is injected.

The multiple-choice query Q is automatically constructed from templates based on the type of the question and the ground-truth event, which makes the task scalable while preserving a fixed decision interface across patients and visits. The three subtasks are defined as follows:

- **Next Action Prediction** asks the agent to predict the future event action type a_i^{GT} from the clinical action space $\mathcal{A}$.
- **Argument Prediction** asks the agent to predict the target event argument p_i^{GT} , such as which laboratory panels or imaging modality are most appropriate.
- **Discharge Decision** asks whether the patient should be discharged within 6 hours, yielding a binary Yes/No decision.

At inference time, the agent receives S , an instructional prompt, M , and Q . The task is formulated as a Long-Horizon Reasoning Task because the model is explicitly required to reason before committing to the next clinical decision. Given the current context and injected historical memory, the model first produces an explicit chain-of-thought-style rationale in the JSON `reason` field and then returns the selected decision in the JSON `answer` field. This reason-then-act interface follows the general design philosophy of ReAct-style agents, where reasoning is used to support action selection [31]. The design also allows for qualitative analysis of the reasoning process, enabling identification of specific failure modes such as retrieval errors, temporal misjudgments, or flawed inference as shown in Appendix A.5.

2.3.2 Evaluation Metrics

Next Action Prediction and Discharge Decision are evaluated by classification accuracy, while Argument Prediction is evaluated by top- k accuracy because multiple arguments may be clinically appropriate. Discharge Decision is evaluated directly as binary classification accuracy. For Next Action Prediction and Argument Prediction, we additionally apply a time-decay scoring mechanism to recognize decisions matching action occurs within 24 hours after the sampled timestamp. This design evaluates whether the

model can make a temporally appropriate decision, rather than whether it can reproduce one exact event index. Let

$$\mathcal{E}_{24h} = \{E_i^k \mid t_i^k \in [t_i^{GT}, t_i^{GT} + 24h]\}.$$

The score is

$$S = \max_{E_i^k \in \mathcal{E}_{24h}} \exp\{-\lambda(t_i^k - t_i^{GT})\} \cdot \text{Score}(y_i^k, \hat{y}), \quad \lambda = \frac{\ln(100)}{24},$$

where the decay factor decreases from 1.0 at the target timestamp to 0.01 after 24 hours, $\text{Score}(\cdot)$ is classification accuracy for Next Action Prediction and top- k accuracy for Argument Prediction.

2.4 Experimental Setup

2.4.1 Models

We evaluate three state-of-the-art LLMs: GPT-5-mini [28], DeepSeek-v3.2 [29], and Qwen-Turbo [30]. All are evaluated through their respective APIs with standardized prompts and default model configurations.

2.4.2 Agent Architectures

Given the fixed memory modules $\{\mathcal{M}_N^{(i,j)}, \mathcal{M}_E^{(i,j)}, \mathcal{M}_C^{(j,T)}\}$, three strategies control how historical memory is provided to the agent: **(1) Long-Context LLM.** $\mathcal{M}_N^{(i,j)}$ or $\mathcal{M}_E^{(i,j)}$ is provided in full within the model’s context window alongside $\mathcal{M}_C^{(j,T)}$, testing native long-context attention without external retrieval assistance. **(2) RAG.** $\mathcal{M}_N^{(i,j)}$ or $\mathcal{M}_E^{(i,j)}$ is externally indexed. At decision time, $\mathcal{M}_C^{(j,T)}$ serves as the retrieval query; top- k relevant chunks are retrieved and concatenated with the context. **(3) Agent Memory System.** We integrate the Mem0 framework [24], providing a persistent, evolvable memory store to the agent. The agent can add, search, and update memory across a patient’s trajectory, actively managing what is stored and retrieved. The design tests whether agentic memory system outperforms passive retrieval for action-oriented rea-

soning.

2.4.3 Experimental Conditions

We design a factorial ablation varying four factors:

1. **Memory Granularity:** Note Memory vs. Event Memory.
2. **Provision Strategy:** Naive Long-Context vs. RAG vs. Mem0.
3. **Historical Depth (n):** Number of injected historical visits: $n \in \{0, 2, 5\}$. $n = 0$ provides Context Memory $\mathcal{M}_C^{(j,T)}$ only with no history as baseline.
4. **Reasoning Mode:** Standard vs. model-native thinking mode (DeepSeek-v3.2 only), where the thinking condition uses DeepSeek’s native deep exploration/reasoning mode rather than an ordinary prompt-level chain-of-thought exemplar.

Qwen-Turbo serves as the primary ablation vehicle; key conditions are replicated on GPT-5-mini and DeepSeek-v3.2 for cross-model validation.

2.5 Results and Analysis

Table 2.1 Long-context LLM performance on the LongMedBench Long-Horizon Reasoning Task. Argument = event argument prediction; Discharge = six-hour discharge decision; Action = next action-type prediction.

Model	Note Memory				Event Memory			
	Argument	Discharge	Action	Avg	Argument	Discharge	Action	Avg
Qwen-Turbo	0.475	0.604	0.309	0.434	0.494	0.604	0.309	0.420
DeepSeek-v3.2	0.478	0.607	0.278	0.422	0.493	0.607	0.304	0.439
GPT-5-mini	0.478	0.625	0.324	0.445	0.494	0.607	0.335	0.452
DeepSeek-v3.2-thinking	0.471	0.578	0.294	0.420	0.466	0.593	0.290	0.419

2.5.1 Model Level Patterns

Table 2.1 reports the long-context LLM performance, where each model is allowed to use the available patient history within a maximum 128K-token context. The strongest overall result is obtained by GPT-5-mini with Event Memory (Avg = 0.452), followed by GPT-5-mini with Note Memory (Avg = 0.445) and DeepSeek-v3.2 with Event Memory

(Avg = 0.439). This ranking supports the LongMedBench observation that structured event memory can outperform note memory for several models in decision-making, but the advantage is not universal: Qwen-Turbo performs better with Note Memory, and DeepSeek-v3.2-thinking remains nearly unchanged across memory types.

At the subtask level, Discharge Decision reaches approximately 0.58–0.63 because it is a binary discharge decision, whereas Next Action Prediction remains around 0.28–0.34 because selecting the next action type requires finer-grained clinical planning. Argument Prediction lies in the middle, reflecting the difficulty of selecting appropriate event arguments such as laboratory panels or imaging targets.

Event Memory slightly improves Argument Prediction in most models, suggesting that fine-grained historical events help parameter selection, but the gains do not consistently translate into higher average decision-making performance. DeepSeek-v3.2’s thinking mode does not improve decision-making performance in this benchmark. Under Event Memory, the average score drops from 0.439 to 0.419, and Discharge Decision decreases from 0.607 to 0.593. This suggests that extended model-native reasoning may make the model more conservative or distract it from committing to the multiple-choice decision, especially when the task requires selecting an actionable next step rather than explaining uncertainty.

Table 2.2 Qwen-Turbo ablation of historical visit injection and retrieval/memory architectures. The upper block varies injected visit depth; the lower block compares RAG and Mem0 over event and note memory.

Setting	Memory/Arch	Argument	Discharge	Action	Avg
Visit Injection					
$n = 0$	–	0.49	0.61	0.32	0.45
$n = 2$	Event	0.49	0.60	0.32	0.44
$n = 5$	Event	0.48	0.60	0.31	0.43
$n = 2$	Note	0.48	0.60	0.32	0.44
$n = 5$	Note	0.48	0.60	0.32	0.44
Memory Architecture					
RAG	Event	0.45	0.51	0.29	0.40
Mem0	Event	0.47	0.62	0.31	0.44
RAG	Note	0.48	0.60	0.31	0.44
Mem0	Note	0.46	0.56	0.29	0.41

2.5.2 Memory Ablations: Visit Injection and Retrieval Architectures

Table 2.2 reports the Qwen-Turbo ablation used to test historical depth and memory-augmentation strategies. Increasing the number of injected visits does not produce monotonic gains, and retrieval-based architectures remain close to or below the naive baseline.

Finding. The ablation results revise the earlier interpretation: the negative result is not specific to Discharge Decision, but spans the broader decision-making suite. Even when historical information is made available through longer visit injection, RAG, or Mem0, models do not systematically improve on Argument Prediction, Discharge Decision, or Next Action Prediction. The $n = 0$ baseline remains competitive, indicating that agents rely heavily on immediate context and struggle to convert retrieved history into better decisions.

Chapter 3. Discussion and Conclusion

3.1 Summary of Findings

The experimental in Chapter 2 showed a consistent pattern: (1) long-horizon clinical reasoning remains challenging, especially for Next Action Prediction; (2) Discharge Decision is easier and should not be used as a proxy for the entire reasoning benchmark; (3) Event Memory can modestly improve Argument Prediction but does not reliably improve average decision-making performance; (4) historical visit injection, RAG, and Mem0 do not provide systematic gains over the immediate-context baseline; and (5) DeepSeek’s model-native thinking mode can degrade performance through increased conservatism.

3.2 The Retrieval-Reasoning Gap

These findings collectively reveal a retrieval-reasoning gap: a disconnect between LLMs’ ability to locate relevant historical information and their ability to integrate that information into proactive clinical planning. RAG and long-term memory systems optimize access to relevant context [20, 21, 25, 32], but for clinical decision-making, retrieval is not always the bottleneck. The model can access relevant history, either in full context or through retrieval, yet still fail to use that history to choose a better action.

This negative result is therefore informative. It suggests that the limitation is not only the amount of available context, but the absence of mechanisms that connect retrieved evidence to the current decision. The relevant target for future work is not simply retrieving more history, but modeling why a prior event matters, whether it changed the patient’s trajectory, and how it should affect the next clinical action. The qualitative traces in Appendix A.5 illustrate these failure modes at the individual-query level.

The contrast with EHR-Bench [12] and recent temporal EHR reasoning work [11, 33] further clarifies the gap. These studies show that LLMs can model clinical trajectories with non-trivial accuracy, but the agent-based setting studied here shows that this ability does not automatically transfer to interactive decision-making, where history

must be selectively retrieved, judged for relevance, and converted into a plan. Static next-event prediction and interactive planning are therefore complementary but distinct capabilities.

We hypothesize three contributing mechanisms. First, clinical reasoning requires causal, not merely associative, inference: determining whether a prior treatment was effective requires understanding why it was given and how the patient responded. Second, longitudinal EHR data have a low signal-to-noise ratio, so a small number of decision-relevant events may be buried in hundreds of routine records. Third, current LLM training data lack the temporal-causal structure of clinical trajectories, creating a distribution shift that retrieval alone cannot remove.

3.3 Benchmark Validity and Clinical Ground Truth

The broader LongMedBench design includes separate retrieval, temporal reconstruction, and reasoning for clinical action. Factual QA primarily tests whether a model can find information in long clinical histories, Temporal Reasoning tests event and visit ordering, and the Long-Horizon Reasoning Task studied here asks whether the model can turn longitudinal evidence into an explicit rationale and then into an action or decision. This separation aligns with emerging virtual EHR and clinical-agent benchmarks that evaluate agent behavior rather than isolated medical knowledge alone [6, 34]. The dissociation across these task families sharpens the central claim: memory augmentation may help retrieval-oriented tasks without reliably improving decision-making.

LongMedBench derives labels from real EHR trajectories, but an observed clinical record is not automatically a perfect ground truth. A subsequent order, medication, or discharge timing reflects the actual care process under local constraints, documentation habits, clinician preferences, resource availability, and possibly clinical error. This limitation is central to clinical decision-making evaluation, where real-world action may not equal to gold answer [35]. The decision task should therefore be understood as evaluation on whether an agent can identify the action most consistent with the observed patient trajectory, not whether it can certify the globally optimal clinical treatment.

In our experiments, several design choices mitigate, but do not eliminate, this limitation. (1) The multiple-choice format uses clinically plausible alternatives, turning evaluation into a context-discrimination problem rather than open-ended treatment generation. (2) The 24-hour time-decay score described in Chapter 2 also rewards temporally proximate matching events instead of forcing exact agreement with a single timestamp. These mechanisms acknowledge that multiple clinically reasonable actions may exist at a given state. Nevertheless, they cannot replace prospective expert adjudication; human clinician review remains necessary to distinguish truly wrong answers from acceptable alternatives.

3.4 Limitations and Future Work

LongMedBench is built from MIMIC-IV, a single-institution database, and the filtered cohort emphasizes high-utilization patients with matched discharge notes. Its event representation reduces noise by focusing on abnormal laboratory results and salient structured events, but this may underrepresent the difficulty of distinguishing normal variation from pathology. Cross-institutional validation on datasets such as eICU [36], HiRID [37], or STARR [38] is therefore needed. Future work should also include dedicated evaluation of medically adapted language models, rather than testing only general-purpose LLMs, to determine whether domain-specific pretraining or clinical instruction tuning improves longitudinal reasoning.

The decision tasks also simplify real clinical practice into automatically templated Next Action Prediction, Argument Prediction, and Discharge Decision queries. This improves diagnostic clarity and scalable construction, but omits factors such as dosage, contraindications, patient preferences, resources, and guideline compliance; Discharge Decision is also weakly discriminative in the sampled evaluation. Future work should add clinician adjudication, adversarial or counterfactual cases, stronger memory architectures for temporal evidence, multimodal data, and outcome-linked evaluation such as mortality, readmission, or length of stay.

3.5 Reproducibility

MIMIC-IV requires credentialed access through PhysioNet, but to support reproducibility and future extensions, we will release the data processing pipeline, template-based task generation logic, and evaluation harness if LongMedBench is admitted by MICCAI 2026. API-based model evaluation used default settings and fixed random seeds where applicable, and the released codebase documents full prompt templates, memory formatting conventions, and decision queries for replication.

3.6 Conclusion

Clinical practice is inherently longitudinal, requiring clinicians to integrate evidence accumulated across years of patient history. This thesis introduced the Long-Horizon Reasoning Task of LongMedBench to evaluate whether contemporary LLM agents can use such historical information through a three-tier memory architecture and systematic ablation experiments.

The central finding is that memory augmentation does not significantly improve longitudinal clinical decision-making: accuracy remains dominated by immediate context, while historical information adds only marginal value. The bottleneck is therefore not just retrieval, but integration. Closing this retrieval-reasoning gap will require architectures that reason over temporal clinical evidence and benchmarks that distinguish empirical EHR trajectories from clinically optimal decisions.

References

- [1] GONG E J, BANG C S, LEE J J, et al. Knowledge-Practice Performance Gap in Clinical Large Language Models: Systematic Review of 39 Benchmarks.[J/OL]. Journal of medical Internet research, 2025, 27: e84120. DOI: 10.2196/84120.
- [2] JOHNSON A, BULGARELLI L, POLLARD T, et al. MIMIC-IV[J/OL]. PhysioNet, 2024. <https://doi.org/10.13026/kpb9-mt58>.
- [3] CAI Y, WANG L, WANG Y, et al. MedBench: A Large-Scale Chinese Benchmark for Evaluating Medical Large Language Models[C/OL]//Proceedings of the AAAI Conference on Artificial Intelligence: Vol. 38. Vancouver, Canada, 2024: 17709-17717. <https://ojs.aaai.org/index.php/AAAI/article/view/29723>. DOI: 10.1609/aaai.v38i16.29723.
- [4] LEE G, HWANG H, BAE S, et al. EHRSQL: A Practical Text-to-SQL Benchmark for Electronic Health Records[C/OL]//Advances in Neural Information Processing Systems: Vol. 35. New Orleans, LA, USA, 2022: 15589-15601. https://papers.nips.cc/paper_files/paper/2022/hash/643e347250cf9289e5a2a6c1ed5ee42e-Abstract-Datasets_and_Benchmarks.html.
- [5] FLEMING S L, LOZANO A, HABERKORN W J, et al. MedAlign: A Clinician-Generated Dataset for Instruction Following with Electronic Medical Records [C/OL]//Proceedings of the AAAI Conference on Artificial Intelligence: Vol. 38. Vancouver, Canada, 2024: 22021-22030. <https://ojs.aaai.org/index.php/AAAI/article/view/30205>. DOI: 10.1609/aaai.v38i20.30205.
- [6] JIANG Y, BLACK K C, GENG G, et al. MedAgentBench: A Virtual EHR Environment to Benchmark Medical LLM Agents[J/OL]. NEJM AI, 2025, 2(9): AIdbp2500144. <https://ai.nejm.org/doi/full/10.1056/AIdbp2500144>.
- [7] SCHMIDGALL S, ZIAEI R, HARRIS C, et al. AgentClinic: a multimodal agent benchmark to evaluate AI in simulated clinical environments[A/OL]. 2025. arXiv: 2405.07960. <https://arxiv.org/abs/2405.07960>.
- [8] MEHANDRU N, GOLCHINI N, ZACK T, et al. ER-REASON: A Benchmark Dataset for LLM-Based Clinical Reasoning in the Emergency Room[A/OL]. 2025. arXiv: 2505.22919. <https://arxiv.org/abs/2505.22919>.
- [9] LEE G, BACHE Y, YANG E, et al. FHIR-AgentBench: Benchmarking LLM Agents for Realistic Interoperable EHR Question Answering[C/OL]//Proceedings of Machine Learning Research: Vol. 297 Proceedings of Machine Learning for Health. 2025. <https://openreview.net/forum?id=Le1hGVQNb8>.
- [10] XU R, ZHUANG Y, ZHONG Y, et al. MedAgentGym: A Scalable Agentic Training Environment for Code-Centric Reasoning in Biomedical Data Science[A/OL]. 2025. arXiv: 2506.04405. <https://arxiv.org/abs/2506.04405>.
- [11] CUI H, UNELL A, CHEN B, et al. TIMER: Temporal Instruction Modeling and Evaluation for Longitudinal Clinical Records[J/OL]. npj Digital Medicine, 2025, 8: 577. <https://www.nature.com/articles/s41746-025-01965-9>. DOI: 10.1038/s41746-025-01965-9.
- [12] LIAO Y, WU C, LIU J, et al. EHR-R1: A Reasoning-Enhanced Foundational Language Model for Electronic Health Record Analysis[A/OL]. 2025. arXiv: 2510.25628. <https://arxiv.org/abs/2510.25628>.
- [13] ZHU W, WANG X, ZHENG H, et al. PromptCBLUE: A Chinese Prompt Tuning Benchmark for the Medical Domain[A/OL]. 2023. arXiv: 2310.14151. <https://arxiv.org/abs/2310.14151>.

- //arxiv.org/abs/2310.14151.
- [14] HSIEH C P, SUN S, KRIMAN S, et al. RULER: What's the Real Context Size of Your Long-Context Language Models?[A/OL]. 2024. arXiv: 2404.06654. <https://arxiv.org/abs/2404.06654>.
 - [15] KOČISKÝ T, SCHWARZ J, BLUNSOM P, et al. The NarrativeQA Reading Comprehension Challenge[A/OL]. 2017. arXiv: 1712.07040. <https://arxiv.org/abs/1712.07040>.
 - [16] YANG Z, QI P, ZHANG S, et al. HotpotQA: A Dataset for Diverse, Explainable Multi-hop Question Answering[C/OL]//Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing. Brussels, Belgium: Association for Computational Linguistics, 2018: 2369-2380. <https://aclanthology.org/D18-1259/>. DOI: 10.18653/v1/D18-1259.
 - [17] MAHARANA A, LEE D H, TULYAKOV S, et al. Evaluating Very Long-Term Conversational Memory of LLM Agents[C/OL]//Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers). Bangkok, Thailand: Association for Computational Linguistics, 2024: 13851-13870. <https://aclanthology.org/2024.acl-long.747/>. DOI: 10.18653/v1/2024.acl-long.747.
 - [18] WU D, WANG H, YU W, et al. LongMemEval: Benchmarking Chat Assistants on Long-Term Interactive Memory[C/OL]//Proceedings of the International Conference on Learning Representations. 2025. https://proceedings.iclr.cc/paper_files/paper/2025/hash/d813d324dbf0598bbdc9c8e79740ed01-Abstract-Conference.html.
 - [19] SHEN Y, HUANG Z, WANG Z, et al. TRIP-Bench: A Benchmark for Long-Horizon Interactive Agents in Real-World Scenarios[A/OL]. 2026. arXiv: 2602.01675. <https://arxiv.org/abs/2602.01675>.
 - [20] ZHAO Y, YUAN B, HUANG J, et al. AMA-Bench: Evaluating long-horizon memory for agentic applications[A/OL]. 2026. arXiv: 2602.22769. <https://arxiv.org/abs/2602.22769>.
 - [21] LEWIS P, PEREZ E, PIKTUS A, et al. Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks[C/OL]//Advances in Neural Information Processing Systems: Vol. 33. Virtual, 2020: 9459-9474. <https://proceedings.neurips.cc/paper/2020/hash/6b493230205f780e1bc26945df7481e5-Abstract.html>.
 - [22] HU Y, LIU S, YUE Y, et al. Memory in the Age of AI Agents[J/OL]. CoRR, 2025, abs/2512.13564. <https://doi.org/10.48550/arXiv.2512.13564>.
 - [23] PACKER C, WOODERS S, LIN K, et al. MemGPT: Towards LLMs as Operating Systems[A/OL]. 2024. arXiv: 2310.08560. <https://arxiv.org/abs/2310.08560>.
 - [24] CHHIKARA P, KHANT D, ARYAN S, et al. Mem0: Building production-ready ai agents with scalable long-term memory[A/OL]. 2025. arXiv: 2504.19413. <https://arxiv.org/abs/2504.19413>.
 - [25] XU W, LIANG Z, MEI K, et al. A-Mem: Agentic memory for llm agents [C/OL]//Advances in Neural Information Processing Systems: Vol. 38. 2025: 17577-17604. https://papers.nips.cc/paper_files/paper/2025/hash/19909c36f51abc4856b4560aff3d36d6-Abstract-Conference.html.
 - [26] HU C, GAO X, ZHOU Z, et al. EverMemOS: A Self-Organizing Memory Operating System for Structured Long-Horizon Reasoning[A/OL]. 2026. arXiv:

- 2601.02163. <https://arxiv.org/abs/2601.02163>.
- [27] ZHANG Y, SHU J, MA Y, et al. Memory as Action: Autonomous Context Curation for Long-Horizon Agentic Tasks[A/OL]. 2026. arXiv: 2510.12635. <https://arxiv.org/abs/2510.12635>.
- [28] OpenAI. GPT-5 mini Model[EB/OL]. 2025. <https://developers.openai.com/api/docs/models/gpt-5-mini>.
- [29] DeepSeek-AI. DeepSeek-V3.2: Pushing the Frontier of Open Large Language Models[A/OL]. 2025. arXiv: 2512.02556. <https://api-docs.deepseek.com/news/news251201>.
- [30] Qwen. Text Generation Models: Qwen-Turbo[EB/OL]. 2026. <https://docs.qwencloud.com/developer-guides/getting-started/text-generation-models>.
- [31] YAO S, ZHAO J, YU D, et al. ReAct: Synergizing Reasoning and Acting in Language Models[A/OL]. 2022. arXiv: 2210.03629. <https://arxiv.org/abs/2210.03629>.
- [32] Mem0. Mem0: Universal memory layer for ai agents[EB/OL]. 2023. <https://github.com/mem0ai/mem0>.
- [33] QU Z, FÄRBER M. TRACE: Temporal Reasoning via Agentic Context Evolution for Streaming Electronic Health Records (EHRs)[A/OL]. 2026. arXiv: 2602.12833. <https://arxiv.org/abs/2602.12833>.
- [34] LIAO Y, XUAN C, CAI Y, et al. AgentEHR: Advancing autonomous clinical decision-making via retrospective summarization[A/OL]. 2026. arXiv: 2601.13918. <https://arxiv.org/abs/2601.13918>.
- [35] HAGER P, JUNGSMANN F, HOLLAND R, et al. Evaluation and Mitigation of the Limitations of Large Language Models in Clinical Decision-Making[J/OL]. *Nature Medicine*, 2024, 30(9): 2613-2622. <https://www.nature.com/articles/s41591-024-03097-1>. DOI: 10.1038/s41591-024-03097-1.
- [36] POLLARD T J, JOHNSON A E W, RAFFA J D, et al. The eICU Collaborative Research Database, a Freely Available Multi-Center Database for Critical Care Research[J/OL]. *Scientific Data*, 2018, 5: 180178. DOI: 10.1038/sdata.2018.178.
- [37] YÈCHE H, KUZNETSOVA R, ZIMMERMANN M, et al. HiRID-ICU-Benchmark — A Comprehensive Machine Learning Benchmark on High-Resolution ICU Data[C/OL]//Thirty-fifth Conference on Neural Information Processing Systems Datasets and Benchmarks Track. Virtual, 2021. <https://arxiv.org/abs/2111.08536>.
- [38] CALLAHAN A, ASHLEY E, DATTA S, et al. The Stanford Medicine Data Science Ecosystem for Clinical and Translational Research[J/OL]. *JAMIA Open*, 2023, 6(3): ooad054. DOI: 10.1093/jamiaopen/oad054.

Appendix

This appendix reorganizes the supplementary material around the benchmark artifacts used to construct and evaluate LongMedBench. Code-like artifacts are shown in shaded blocks with a colored header. Long notes, reports, and model responses are shortened only for readability while preserving the original schema and decision content.

A1 Dataset Pipeline

The dataset pipeline starts from raw MIMIC-IV notes and structured tables, converts each admission into a typed clinical event stream, and then derives the current-visit interaction context used by the agentic decision task. A representative patient-level sequence record contains 813 ordered events for Patient P000001, including demographics, admissions, labs, microbiology, imaging, medications, procedures, and discharges.

A1.1 Raw Discharge Note Example

The raw note table stores one discharge summary per row. A representative row has the schema `note_id`, `subject_id`, `hadm_id`, `note_type`, `note_seq`, `charttime`, `storetime`, and `text`.

Raw Discharge Note Row

```
note_id: 10000032-DS-21
subject_id: 10000032
hadm_id: 22595853
note_type: DS
charttime: 2180-05-07 00:00:00
storetime: 2180-05-09 15:26:00
text: Name: ___ Unit No: ___
Admission Date: ___ Discharge Date: ___
Service: MEDICINE
Allergies: No Known Allergies / Adverse Drug Reactions
Chief Complaint: Worsening ABD distension and pain
Major Surgical or Invasive Procedure: Paracentesis
History of Present Illness: ___ HCV cirrhosis c/b ascites,
hiv on ART, h/o IVDU, COPD, bipolar, PTSD, presented from OSH ED
```

```
with worsening abdominal distension and pain ...
```

During preprocessing, admission-facing fields such as chief complaint and history of present illness initialize the current visit, while discharge-facing fields become visit-level Note Memory for later visits.

A1.2 Structured Admission and Discharge Events

In the sequenced patient file, a visit is represented by explicit boundary events. For example, the first retained visit of P000001 begins with an admission event and ends with a discharge event.

Structured Admission and Discharge Events

```
{
  "event_id": "P000001-V1-adm",
  "event_type": "ADMISSION",
  "timestamp": "2120-08-06 20:15:00",
  "visit_ref": "V1",
  "content": {
    "location": "PACU",
    "admission_type": "EW EMER.",
    "chief_complaint": "hypotension and lethargy s/p RFA",
    "history_of_present_illness": "Mr. ___ is a ___ year old man ...",
    "insurance": "Medicaid"
  }
}

{
  "event_id": "P000001-V1-dis",
  "event_type": "DISCHARGE",
  "timestamp": "2120-08-14 19:30:00",
  "visit_ref": "V1",
  "content": {
    "discharge_time": "2120-08-14 19:30:00",
    "discharge_location": "HOME",
    "discharge_note": {...}
  }
}
```


A1.3 Radiology and Imaging Event Example

Radiology reports are converted into imaging events with a timestamp and report text. The same patient sequence contains the following imaging event.

Radiology to Imaging Event

```
{
  "event_id": "P000001-V1-E2",
  "event_type": "IMAGING",
  "timestamp": "2120-08-06 07:03:00",
  "visit_ref": "V1",
  "content": {
    "timestamp": "2120-08-06 07:03:00",
    "report": "CONTRAST-ENHANCED CT OF LIVER AND UPPER ABDOMEN\n\n"
              "INDICATION: ___ male with known cirrhosis and abnormalities "
              "in segment II and segment VI of the right lobe of liver. ..."
```

For Context Memory, this event is rendered as an assistant action such as `order_imaging` followed by an environment observation containing the available radiology result.

A2 Memory Construction

LongMedBench uses three memory forms. Note Memory stores previous-visit discharge summaries, Event Memory stores previous-visit structured event streams, and Context Memory stores the dialogue-style state of the current visit before a decision point.

A2.1 LLM Convert

Context Memory is the only memory type that uses an LLM during construction. The code first maps structured events to semantic actions, then asks the LLM to write the assistant-side rationale field under strict grounding constraints.

ContextMemory LLM Conversion Prompt

You write the 'reason' field for the next assistant action in a doctor-agent dialogue.

Hard constraints:

- You may ONLY use information present in the provided prefix_messages.
- Do NOT invent new patient facts, findings, diagnoses, or outcomes.
- Do NOT mention results of the current action (results are not available yet).
- Keep it 1-2 sentences, not exceeding 50 words, workflow-rationale focused.

Output ONLY JSON: {"reason": string, "constraints_ok": boolean}.

User payload:

```
{
  "prefix_messages": <system, admission, and previous interaction messages>,
  "current_decision": {"action": <action>, "args": <arguments>}
}
```

The remaining memory types are constructed automatically. Note Memory is produced by extracting the complete discharge-note text from previous visits. Event Memory is produced by chronological serialization of structured EHR events. Neither requires LLM rewriting.

A2.2 Memory Templates

NoteMemory Template

[Visit P000001-V1 NoteMemory]

Chief Complaint: hypotension and lethargy s/p RFA

History of Present Illness: Mr. ___ is a ___ year old man with

EtOH and HCV cirrhosis, c/b ascites, prior variceal bleeds, recent MRI with 2 lesions suspicious for ___, who was admitted s/p RFA with lethargy and hypotension. ...

Discharge Location: HOME

Discharge Summary: <complete discharge-note fields from raw note>

EventMemory Template

```

PROCEDURE P000001-V1-E1 2120-08-06
  items: Closed percutaneous needle biopsy of liver;
        Percutaneous ablation of liver lesion or tissue

LAB P000001-V1-E8 2120-08-08 05:10:00
  items: Red Blood Cells 3.29 m/uL abnormal; RDW 16.7% abnormal;
        Alanine Aminotransferase included in Chemistry panel

MICRO P000001-V1-E11 2120-08-08 11:45:00
  specimen: BLOOD CULTURE; result: NO GROWTH

MEDICATION P000001-V1-E4 2120-08-06 21:00:00
  items: Morphine Sulfate IV 0.5-1 mg ordered;
        Ondansetron IV 4 mg ordered

```

ContextMemory Template

```

{
  "patient_id": "P000001",
  "visit_id": "P000001-V12",
  "messages": [
    {"role": "system", "content": "You are a doctor agent ..."},
    {"role": "user", "content": "Gender: M. Age at first visit: 48 ..."},
    {"role": "assistant", "content": {
      "reason": "The patient has atypical chest pain and bilateral lower
                extremity edema, warranting a chest X-ray ...",
      "action": "order_imaging",
      "args": {"modality": "xray", "target": "chest",
               "timestamp": "2121-12-16 19:05:00"}
    }},
    {"role": "user", "content": {
      "name": "environment",
      "observation_type": "imaging_results",
      "results": [{"test": "chest xray", "content": "..."}]
    }}
  ]
}

```

A3 Question Generation

The agentic decision task uses three question templates generated from the current event prefix and future event stream. The examples below are representative generated questions for Patient P000001.

Table 3.1 Question templates for the three agentic decision subtasks.

Type	Template	Example options	Answer form
T3-N Next Action	Which of the next step is most appropriate?	perform_procedure, medication, ask_question, order_microbiology, order_labs, order_imaging	Weighted action dictionary
T3-A Argument	Which imaging order is most appropriate next?	CT head, MRI brain, CT abdomen, XRAY abdomen, CT chest, US pelvis	Weighted argument dictionary
T3-D Discharge	Should the patient be discharged within 6h?	Yes, No	Weighted binary dictionary

Expected Model Output

```
{"reason": "The patient has abnormal lab results...", "answer": ["order_labs"]}
```

For T3-A questions, candidate answers are collected from future events in a 24-hour window after the current timestamp and weighted by temporal decay. T3-D instead compares the current timestamp with the discharge time using a six-hour horizon.

A4 Agent Configuration

Each evaluation condition is defined by two axes: the historical memory source, either Note Memory or Event Memory, and the delivery strategy, either long-context prompting, retrieval-augmented generation (RAG), or Mem0-style managed memory. The current-visit Context Memory is always included as the immediate interaction state.

A5 Case Study

This section provides concrete qualitative cases that explain why aggregate accuracy changes are small even when memory configuration changes individual decisions. Each case is centered on the original interaction-style JSON. Explanatory text is included only to summarize the behavioral pattern illustrated by the JSON.

Table 3.2 Agent context structure under each memory-delivery setting.

Setting	Historical source	Context structure
Note + context	Previous discharge notes	System prompt, current-visit context messages, packed previous-visit discharge notes, user question.
Event + context	Previous event streams	System prompt, current-visit context messages, packed chronological events from previous visits, user question.
Note + RAG	Previous discharge notes indexed for retrieval	System prompt and current context, retrieved note snippets relevant to the question, user question.
Event + RAG	Previous event streams indexed for retrieval	System prompt and current context, retrieved event snippets relevant to the question, user question.
Note + Mem0	Previous discharge notes stored as managed memories	System prompt and current context, Mem0-recalled note memories, user question.
Event + Mem0	Previous event streams stored as managed memories	System prompt and current context, Mem0-recalled event memories, user question.

A read-only inspection of the analysis artifacts showed that event and note memories often change model outputs even when the overall benchmark average remains close. In the shared 2,202-question overlap, event–note prediction changes occurred in 662 questions for the long-context agent (30.06%), 807 questions for Mem0 (36.65%), and 740 questions for RAG (33.61%). The cases below therefore focus on how those changes arise rather than treating the mean score as the only evidence.

Case 1: Retrieval helps when local evidence reinforces the current trajectory.

Original Interaction JSON: P000001-V12-T0002-T3-N-0

```
{
  "qid": "P000001-V12-T0002-T3-N-0",
  "qtype": "T3-N",
  "timestamp": "2121-12-16 19:10:00",
  "question": "Which of the next step is most appropriate?",
  "options": [
    "perform_procedure",
    "medication",
    "ask_question",
    "order_microbiology",
    "order_labs",
    "order_imaging"
  ],
  "answer": {"order_imaging": 1.0},
  "context_messages": [
    {
```

```

    "role": "system",
    "content": "You are a doctor agent. You can perform these semantic actions:
ask_question, order_labs, order_microbiology, order_imaging, perform_procedure,
medication, discharge. The environment returns only information that exists in this
admission record; otherwise it returns not_available. Do not invent new findings."
  },
  {
    "role": "user",
    "content": "Gender: M. Age at first visit: 48. Chief complaint: Chest Pa. HPI: male
with HCV and EtOH cirrhosis on the transplant list, complicated by HCC s/p TACE/RFA and
ascites, recurrent C. difficile colitis, increasing abdominal distention, bilateral
lower extremity swelling, and occasional non-pleuritic, non-exertional chest pain for 3
days."
  },
  {
    "role": "assistant",
    "content": {
      "reason": "Atypical chest pain and bilateral lower extremity edema warrant chest X-
ray evaluation.",
      "action": "order_imaging",
      "args": {"modality": "xray", "target": "chest", "timestamp": "2121-12-16 19:05:00"}
    }
  },
  {"role": "environment", "content": "chest xray -> Limited, negative."},
  {
    "role": "assistant",
    "content": {
      "reason": "Bilateral lower extremity edema warrants Doppler ultrasound to evaluate
for deep vein thrombosis.",
      "action": "order_imaging",
      "args": {"modality": "doppler", "target": "free text body part", "timestamp":
"2121-12-16 19:10:00"}
    }
  },
  {"role": "environment", "content": "free text body part doppler -> Addendum: A limited
Color Doppler and Spectral Analysis Study was also performed."}
],
"predictions": {
  "note_context": {"answer": ["order_labs"], "score": 0.0},
  "event_context": {"answer": ["order_labs"], "score": 0.0},
  "note_rag": {"answer": ["order_imaging"], "score": 1.0},

```

```

    "event_rag": {"answer": ["order_imaging"], "score": 1.0},
    "note_mem0": {"answer": ["order_labs"], "score": 0.0},
    "event_mem0": {"answer": ["order_labs"], "score": 0.0}
  }
}

```

RAG succeeds because the retrieved material reinforces the current imaging trajectory, while Mem0 retrieves clinically plausible but temporally mismatched memories that answer a different situation.

Case 2: More memory is not always better.

Original Interaction JSON: P000003-V16-T0009-T3-N-0

```

{
  "memory_type": "event",
  "qid": "P000003-V16-T0009-T3-N-0",
  "qtype": "T3-N",
  "question": "Which of the next step is most appropriate?",
  "gt": ["medication"],
  "baseline": ["order_labs"],
  "baseline_reason": "The patient is undergoing R-ICE chemotherapy and has abnormal lab
    results, indicating the need for further laboratory testing to monitor for potential
    complications.",
  "preds": {
    "ctx0": ["medication"],
    "ctx2": ["medication"],
    "ctx5": ["order_labs"]
  },
  "reasons": {
    "ctx0": "The patient has abnormal lab results and is receiving chemotherapy, so ordering
      furosemide is necessary to manage potential fluid overload or related complications.",
    "ctx2": "The patient has abnormal lab results and is receiving chemotherapy, so ordering
      furosemide is necessary to manage potential fluid overload or related complications.",
    "ctx5": "The patient is undergoing R-ICE chemotherapy and has abnormal lab results,
      indicating the need for further laboratory testing to monitor for potential
      complications."
  }
}

```

The zero-visit and two-visit contexts choose the correct action, but the five-visit context shifts back to laboratory monitoring. This is a non-monotonic context-length effect: adding history can dilute the current action signal rather than clarify it.

Case 3: Event memory can preserve action-relevant signals.

Original Interaction JSON: P000002-V15-T0031-T3-N-0

```
{
  "agent": "agentic_decision",
  "qid": "P000002-V15-T0031-T3-N-0",
  "qtype": "T3-N",
  "question": "Which of the next step is most appropriate?",
  "event_pred": ["medication"],
  "note_pred": ["order_labs"],
  "event_reason": "The patient has abnormal lab results and a positive respiratory culture,
    requiring targeted antibiotic therapy based on susceptibility results. Therefore, the
    most appropriate next step is to administer medication.",
  "note_reason": "The patient has multiple abnormal lab results and a positive respiratory
    culture, indicating the need for further diagnostic evaluation and targeted treatment.
    Ordering labs will help guide management.",
  "gt": ["medication"]
}
```

Both memories recognize abnormal labs and a positive culture, but event memory converts that evidence into the next intervention whereas note memory remains at diagnostic evaluation.

Case 4: Retrieved memory can override the current clinical target.

Original Interaction JSON: P000001-V12-T0002-T3-A-1

```
{
  "memory_type": "note",
  "qid": "P000001-V12-T0002-T3-A-1",
  "qtype": "T3-A",
  "question": "Which imaging order is most appropriate next?",
  "gt": ["XRAY abdomen"],
  "baseline": ["XRAY abdomen"],
```



```

"baseline_reason": "The patient has bilateral lower extremity edema, which is more
                    consistent with a systemic condition such as liver disease or heart failure rather than
                    a localized issue in the extremities. Therefore, an imaging study of the abdomen would
                    be more appropriate to evaluate for underlying causes such as ascites or liver pathology
                    .",
"preds": {
  "llm_agent": ["XRAY abdomen"],
  "memO_agent": ["CT head"],
  "rag_agent": ["CT chest"]
},
"reasons": {
  "llm_agent": "The patient has bilateral lower extremity edema, which is more consistent
                with a systemic condition such as liver disease or heart failure rather than a localized
                issue in the extremities. Therefore, an imaging study of the abdomen would be more
                appropriate to evaluate for underlying causes such as ascites or liver pathology.",
  "memO_agent": "Patient has a reported headache during this admission and a noncontrast
                CT head is the most appropriate initial, rapid study to evaluate acute intracranial
                causes. Other abdominal and chest imaging have recently been performed and are less
                relevant to the new symptom.",
  "rag_agent": "The patient has chest pain with a limited/negative chest x-ray and a
                nondiagnostic/limited lower-extremity Doppler; the next appropriate study to evaluate
                for pulmonary embolism or other intrathoracic causes is a CT chest."
}
}

```

Mem0 and RAG retrieve recognizable clinical signals, but those signals redirect the answer away from the abdominal ascites and liver-pathology trajectory that supports the ground-truth order.

Case 5: Retrieval can correct an immediate-context error.

Original Interaction JSON: P000001-V20-T0002-T3-N-0

```

{
  "memory_type": "note",
  "qid": "P000001-V20-T0002-T3-N-0",
  "qtype": "T3-N",
  "question": "Which of the next step is most appropriate?",
  "gt": ["order_labs"],
  "baseline": ["order_imaging"],

```

```

"baseline_reason": "The patient has a history of HCC and cirrhosis, and the current
  presentation includes right upper quadrant pain and abdominal distention. An ultrasound
  of the right upper quadrant is needed to evaluate for possible complications.",
"preds": {
  "llm_agent": ["order_imaging"],
  "memO_agent": ["order_labs"],
  "rag_agent": ["order_labs"]
},
"reasons": {
  "llm_agent": "The patient has a history of HCC and cirrhosis, and the current
    presentation includes right upper quadrant pain and abdominal distention. An ultrasound
    of the right upper quadrant is needed to evaluate for possible complications.",
  "memO_agent": "RUQ ultrasound excluded biliary obstruction but the patient has cirrhosis
    , ascites, and was being evaluated for infection; laboratory studies are needed to
    assess for leukocytosis, renal/hepatic function, and guide further management.",
  "rag_agent": "Right upper quadrant ultrasound excluded biliary obstruction/
    choledocholithiasis. The patient with cirrhosis, splenomegaly, and perihepatic ascites
    needs laboratory evaluation to assess for infection, hepatic decompensation, and guide
    need for paracentesis or other interventions."
}
}

```

This positive case shows that memory can help when it supplies a completed prior diagnostic step that changes the next action.

Case 6: Note memory can over-emphasize narrative diagnosis.

Original Interaction JSON: P000007-V11-T0033-T3-N-0

```

{
  "agent": "memO_agent",
  "qid": "P000007-V11-T0033-T3-N-0",
  "qtype": "T3-N",
  "question": "Which of the next step is most appropriate?",
  "event_pred": ["order_labs"],
  "note_pred": ["medication"],
  "event_reason": "The patient has persistent severe electrolyte abnormalities (hypokalemia,
    hypocalcemia, hypomagnesemia), metabolic acidosis and rising creatinine despite ongoing
    replacement; repeating labs/ABG and electrolytes is the next appropriate step to assess
    response and guide further therapy.",

```

```

"note_reason": "The patient has a stool culture positive for Salmonella species and
ongoing severe electrolyte abnormalities requiring continued IV/PO repletion and
initiation/adjustment of targeted antimicrobial therapy in an immunocompromised host.
Thus the immediate next step is medication management.",
"baseline": {
  "event": ["order_labs"],
  "note": ["order_labs"]
},
"gt": ["order_labs"],
"event_hit_gt": true,
"note_hit_gt": false
}

```

The note-memory answer foregrounds a high-level infectious diagnosis, whereas the event-memory answer preserves the immediate monitoring need caused by severe electrolyte and acid-base abnormalities.

Table 3.3 Qualitative patterns illustrated by the A5 cases.

Case	Ground truth	Main contrast	Interpretation
P000001-V12-T0002-T3-N-0	order_imaging	RAG correct; context and Mem0 incorrect	Retrieval helps when it reinforces the current imaging trajectory.
P000003-V16-T0009-T3-N-0	medication	ctx0/ctx2 correct; ctx5 incorrect	Longer history can dilute the action signal.
P000002-V15-T0031-T3-N-0	medication	event correct; note incorrect	Event memory can preserve action-level evidence better than narrative notes.
P000001-V12-T0002-T3-A-1	XRAY abdomen	baseline correct; Mem0/RAG incorrect	Retrieved memories can be plausible but target the wrong clinical thread.
P000001-V20-T0002-T3-N-0	order_labs	baseline incorrect; Mem0/RAG correct	Memory helps when prior completed workup changes the next action.
P000007-V11-T0033-T3-N-0	order_labs	event correct; note incorrect	Notes may over-emphasize diagnosis while events preserve monitoring needs.

Taken together, these cases support the retrieval–reasoning gap discussed in Chapter 3. Memory changes the information available to the agent, but the model still has to decide whether that information is current, action-relevant, and causally connected to the next clinical step. The same mechanism can therefore improve one instance, harm another, and leave the aggregate average nearly unchanged.

Author's Biography

1. Personal Information

Name: Zihan Xu

Gender: Male

Date of Birth: February 27, 2004

Email: zihan1.22@intl.zju.edu.cn

2. Education

September 2022–June 2026: Zhejiang University, ZJU-UIUC Institute, Bachelor of Engineering in Computer Engineering.

August 2024–May 2025: University of Illinois at Urbana-Champaign, exchange program in Computer Engineering.

3. Research and Project Experience

Zihan Xu is currently conducting research at the ZJU-AI4H Lab under the supervision of Professor ZuoZhu Liu. His academic interests focus on large language models, agent memory systems, and medical reasoning evaluation.

From June 2025 to September 2025, he interned at the Innovation Center of Yangtze River Delta, Zhejiang University in Jiashan, contributing to projects on agent personas, dynamic personality adaptation, and LLM-based reminiscence therapy systems.

4. Publications and Patents

LongMedBench: Evaluating Medical Agents in Long-Horizon Clinical Reasoning and Decision Making, MICCAI 2026 in-progress.

Appendix III

Task Plan of Thesis Project

Project Title:	Medical Reasoning with Memory-Augmented Agentic System		
(From Oct 18th 2025 to May 22nd 2026) Progress Timeline:	Milestone	Date	Description
	Project Initialization	Oct 18 - Nov 15, 2025	Confirm thesis topic and initial project scope definition.
	Literature Review	Nov 16, 2025 - Jan 7, 2026	Complete a comprehensive review of medical datasets, LLM-based agents, and memory-augmented frameworks.
	Proposal Preparation	Jan 8 - Jan 23, 2026	Prepare the thesis proposal.
	Prototype Development	Jan 24 - Mar 20, 2026	Design the modular architecture integrating dynamic memory retrieval, case-based reasoning, and LLM inference. Establish testing environment.
	Mid-Term Check-Up	Mar 27, 2026	Demonstrate preliminary prototype and discuss methodology improvements.
	Experimental Implementation & Analysis	Mar 28 – May 10, 2026	Conduct experiments on benchmark datasets, evaluating results and refining architecture.
	Thesis Draft Preparation	May 11 – May 18, 2026	Compile results and finalize the full thesis draft for review.
Task details of thesis project:	Comprehensive Literature Review Review research on memory-augmented reasoning systems and clinical decision-support frameworks; identify existing limitations in reasoning continuity and interpretability. Prototype Development Build a functional prototype implementing core reasoning loops and memory components, integrating large language models with dynamic memory retrieval and case-based reasoning modules to simulate clinical diagnostic reasoning. Experimental Evaluation Design experiments to test the system’s diagnostic reasoning accuracy, temporal coherence, and explanation quality compared with baseline LLM approaches.		
Supervisor Name & Signature:	Zuozhu Liu 		Date: Oct 31, 2025

Appendix VIII

On-site Minutes of Oral Defense

Project Title					
Student Name:		ZJU ID		Major	
Supervisor			Academic Title		
Co-supervisor			Academic Title		
Defense Time			Defense Location		
Thesis Committee Members					
Minutes of Oral Defense: (Specific records and evaluations are required for the presentation and answering of questions during the defense)					
Name and Signature of minute recorder:			Chairman of thesis committee & Signature:		
Date:			Date:		

Appendix IX

Assessment Form of Senior Thesis

Supervisor's comments (more than 50 EN words):						
Supervisor's Signature:				Date:		
Thesis Committee check the following box: ☐ Agree to oral defense. ☐ Agree to oral defense after revisions. ☐ Does not meet the requirements for oral defense.						
Name & Signature of committee chair:				Date:		
Comments from Thesis Committee:						
Marking rubrics	Project Proposal (10%)	Mid-term Check (5%)	Senior thesis (50%)	Oral defense (30 %)	Outcome * (5%)	Total (max. 100 marks)
Grades						
Committee Chair's Signature:				Date:		
*Outcome: The detail will be announced later.						